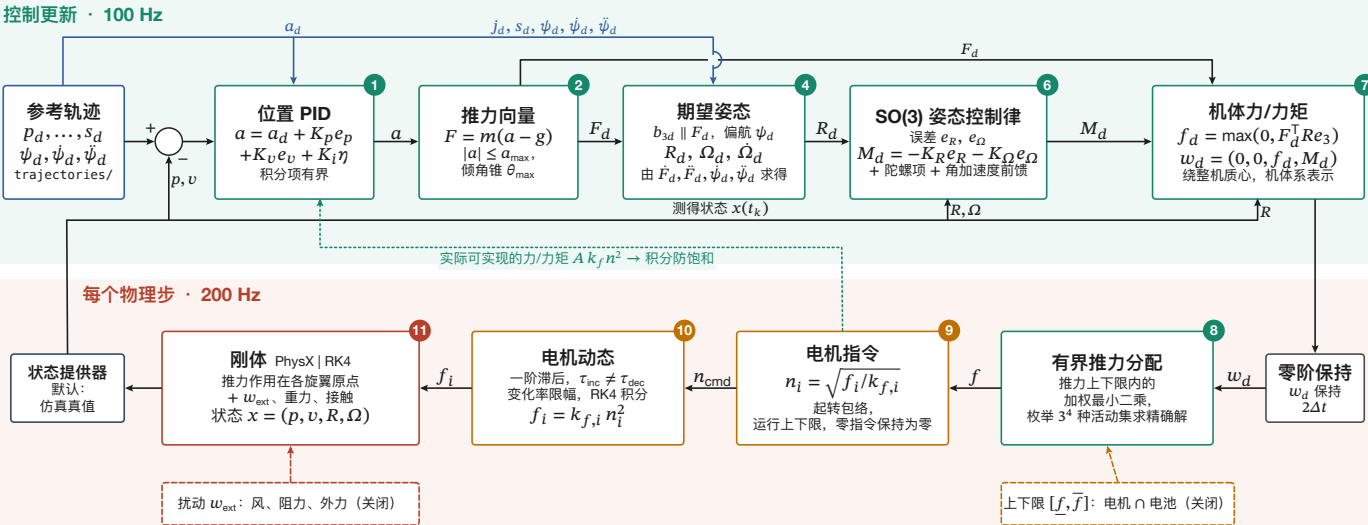


a

几何控制器的闭环信号链 controller: {kind: geometric}; 圆形编号对应 **b** 中的公式

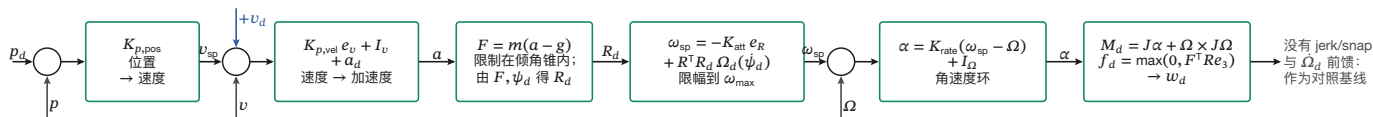
控制更新 · 100 Hz



b 代码中实际使用的公式 $e_p = p_d - p$, $e_v = v_d - v$; $g = (0, 0, -9.81) \text{ m/s}^2$; $h(\psi) = (\cos \psi, \sin \psi, 0)$

- | | | |
|----|---|--|
| 1 | $a = a_d + K_p e_p + K_v e_v + K_I \eta, \quad \dot{\eta} = e_p, \eta_j \leq \eta_{\max,j}$ | geometric.py |
| 2 | $F_d = \Pi_{\theta_{\max}}[m(\text{sata}_{\max}(a) - g)]$ (向倾角锥做正交投影) | shaping.py |
| 3 | $\dot{F}_d = m \dot{J}_d + \frac{d}{dt} F_{fb}, \quad \ddot{F}_d = m \dot{S}_d + \frac{d^2}{dt^2} F_{fb}, \quad F_{fb} = F_d - m a_d$ | geometric.py |
| 4 | $b_{3d} = \frac{F_d}{\ F_d\ }, \quad b_{2d} = \frac{b_{3d} \times h(\psi_d)}{\ b_{3d} \times h(\psi_d)\ }, \quad b_{1d} = b_{2d} \times b_{3d}, \quad \hat{\Omega}_d = R_d^T \dot{R}_d$ | rotations.py |
| 5 | $e_R = \frac{1}{2}(R_d^T R - R^T R_d)^V, \quad e_\Omega = \Omega - R^T R_d \Omega_d$ | geometric.py |
| 6 | $M_d = -K_R e_R - K_\Omega e_\Omega + \Omega \times J \Omega + J(R^T R_d \hat{\Omega}_d - \Omega \times R^T R_d \Omega_d)$ | geometric.py |
| 7 | $f_d = \max(0, F_d^T R e_3), \quad w_d = (0, 0, f_d, M_d)$ | geometric.py |
| 8 | $f = \arg \min_{\underline{f} \leq f \leq \bar{f}} \ W(Af - w_d)\ ^2 + \lambda \ f - f_{\text{prev}}\ ^2$ | allocation.py (枚举 3^4 种活动集) |
| 9 | $n_{\text{cmd},i} = \text{clip}(S(t/T_{\text{su}})\sqrt{f_i/k_{f,i}}, n_{\min,i}, n_{\max,i}), \quad f_i = 0 \text{ 时为 } 0$ | loop.py (起转结束后 $S = 1$) |
| 10 | $\dot{n}_i = \text{clip}(\alpha_i(n_{\text{cmd},i} - n_i), \pm\epsilon), \quad \alpha_i = 1/(\Delta t + \tau_i); \quad n_{\text{cmd},i} < n_i \text{ 时 } \tau_i = \tau_{\text{dec}}$ | actuation.py (否则取 τ_{inc} ; RK4; $n_i \in [0, n_{\max}]$) |
| 11 | $f_i = k_{f,i} n_i^2, \quad A_i = [a_i; (r_i - c) \times a_i + \sigma_i k_\tau a_i], \quad w = Af + w_{\text{ext}}$ | 后端 |

C 级联 PID 插件 controller: {kind: cascaded_pid}: 输入与输出 w_d 均与几何控制器相同; 增益即各环带宽 $[s^{-1}]$



积分项 I_n, I_Ω 有上限, 推力向量或推力分配饱和时停止增长。角速度环按完整惯量 J 归一化, 因此增益可直接理解为与机体尺寸无关的各环带宽。